

Airbnb assignment — how we differentiate

Reading the syllabus as intelligence, and five findings already tested against the data

Every group has the same four files, the same brief, and the same models. The default output is predictable: median price by neighbourhood, room-type splits, a price map, biggest hosts, a correlation heatmap. If we compete on that we're competing on slide design, and the grading is explicitly comparison-based. So the differentiation has to come from somewhere the default doesn't reach.

There are three such places, and they're the structure of this note: **read the syllabus as strategic information, test our own claims and report what fails, and bring data that isn't in the CSVs.**

Worth naming the failure mode we're exploiting. AI-written analysis is fluent, confident, and occasionally wrong in ways that look fine on a slide. The first version of this very plan recommended a distance-from-city-centre analysis that turns out to be backwards in Rio (§6). Most decks in that room will carry at least one confident error like that. A deck that visibly stress-tests itself and reports what broke is very hard to produce by default, and it will read as the only one in the room that checked.

1 What the syllabus tells us

The reference list isn't decoration. Four entries are unusually specific and two of them are close to instructions.

Ehrenberg (1963), "Bivariate regression analysis is useless." He has assigned a paper arguing against the exact technique most groups will build on. Ehrenberg's position is that fitting a line to one bivariate dataset tells you very little, and that what counts is relationships which replicate across many conditions. So: don't lead with a regression. Lead with regularities that hold across all four cities and hundreds of neighbourhoods. Findings 1 and 2 below are built this way. It's a short paper, read it before citing it.

Silberzahn et al. (2018), "Many analysts, one data set." Many independent teams were given identical data and one question, and reached materially different conclusions depending on their analytic choices. Now look at our assignment: roughly fifteen groups, the same four files, graded by comparison. **He is running this paper on the class.** Finding 3 is what we do about it.

Palmatier & Sridhar (2017), Marketing strategy: based on first principles. Compare the book's four principles against the course blocks. Block I is the resource allocation framework and sustainable competitive advantage (*all resources are limited, all competitors react*); Block II is customer heterogeneity and segmentation (*all customers differ*); Block III is customer dynamics (*all customers change*). The course is built on this book. Structuring the deck on the four principles instead of on EDA sections means presenting in his vocabulary rather than ours.

Delen & Zolbanin (2018) gives the descriptive / predictive / prescriptive split. Useful for stating plainly that this project is descriptive and saying what we'd need to go further.

Two practical notes. Gretl is the required software, and blocks V–VII are descriptive statistics, probability and bivariate analysis. That fixes the technical register: a deck with k-means and random forests is off-syllabus, while descriptive and bivariate work done unusually carefully is exactly on it. Reproducing at least the headline result in Gretl and showing its output is cheap and signals we followed the course.

One more thing. The instructor is Brazilian. Rio is the dataset he will read hardest and where a wrong claim gets caught fastest, which cuts both ways: it's also where being right is worth the most.

2 The spine: four first principles, not six EDA blocks

Principle	What we can actually show	Evidence
All customers differ	Heterogeneity is lawlike, not incidental — the entire-home premium never fails	Finding 1
All customers change	We have one snapshot and cannot observe dynamics. Say so directly; proxy it weakly with reviews/month against cumulative reviews	limitation
All competitors react	Four markets, four different professional pricing strategies	Findings 3 & 4
All resources are limited	Where to allocate: price level against demand intensity against regulatory exposure	\$5 + comparison table

Being explicit that principle 2 is unavailable in a cross-section is a strength, not a gap. It shows we know what the data can't do.

3 Five things already tested

All numbers below are computed from the CSVs. Nothing here is a plan; it's done, and it's what the deck should be built around.

Finding 1 — a regularity that replicates in every case (Ehrenberg's method)

Entire homes cost more than private rooms in **100% of the 142 neighbourhoods** with at least 30 of each, across all four cities. The direction never fails once. The magnitude varies systematically by market:

Median across neighbourhoods	Antwerp	Amsterdam	LA	Rio
Entire home ÷ private room	1.57×	1.83×	2.05×	2.49×

The direction is a law; the multiple is a market characteristic. That's a considerably stronger claim than a regression coefficient from one city, and it's the form of evidence Ehrenberg argued for.

Finding 2 — a second replication

Within a given neighbourhood and room type, higher-priced listings get fewer reviews per month. The correlation is negative in 85% of Antwerp cells, 72% of Amsterdam, 98% of Rio and 80% of LA — 250 cells in total. Magnitudes are modest (median r from -0.06 to -0.20). Present it as a replicated regularity, not as a demand curve: we cannot estimate elasticity from this and shouldn't imply we can.

Finding 3 — the centrepiece: the answer depends on choices nobody would call unreasonable

Question: *do professional hosts charge more?* We ran 54 defensible specifications per city — professional threshold 2+/3+/5+ listings, sample all/active-only/entire-homes-only, outliers none/winsorised/trimmed, statistic median/geometric mean.

City	Lowest spec	Highest spec	Median spec	Positive in	Verdict
Antwerp	+12.3%	+73.9%	+27.8%	54/54	stable
Amsterdam	−10.2%	+35.0%	+8.5%	37/54	sign flips
Rio	−27.6%	+19.2%	−7.7%	18/54	sign flips
LA County	−9.7%	+11.3%	−1.2%	22/54	sign flips

In three of four cities the **sign** flips depending on choices any of us would have made without thinking twice. One group will report that Rio professionals charge 19% more; another will report they charge 28% less. Both will be honest, both will show one number, and neither will know the other exists. That is Silberzahn, reproduced on the assigned dataset, in this classroom.

Finding 4 — control properly and four distinct strategies appear

Comparing only listings in the **same neighbourhood and the same room type** (paired cells with at least 15 of each side):

City	Cells	Median premium	Pros cheaper in	Reading
Antwerp	13	+30.0%	1 of 13	Clear premium positioning
Amsterdam	29	+15.7%	6 of 29	Premium — but pooled it looks like +6%
Rio	56	−14.7%	36 of 56	Professionals systematically undercut
LA County	146	+1.1%	73 of 146	A coin flip. A genuine null

Amsterdam is worth naming out loud as **Simpson's paradox**: professionals there hold far more private rooms than amateurs (54% entire homes against 80%), so pooling the room types drags their apparent premium from +15.7% down to about +6%. The composition hides the effect. Rio is the opposite case — the discount is not composition, it survives the control, and professionals there are running a volume strategy rather than a premium one. LA at 73 of 146 cells is as clean a null result as anyone will present.

This is the marketing conclusion and it is what we lead with. Four markets, four competitive postures. Finding 3 is the evidence that the simple version of the question has no single answer; it is not the headline.

Finding 5 — a test that failed, which we report anyway

Double jeopardy is Ehrenberg's own law: smaller brands have fewer buyers *and* those buyers are slightly less loyal. Treating neighbourhoods as brands, it does not replicate here. The correlation between neighbourhood size and median reviews per month is +0.44 in Antwerp, −0.09 in Amsterdam, +0.65 in Rio and −0.21 in LA. No pattern. Reporting a failed test costs us nothing, demonstrates learning outcome 3 directly, and is close to impossible to fabricate.

One data-quality catch worth its own slide. In Catete (Rio), professional entire-home listings have a median of R\$7,269 against R\$228 for amateurs in the same neighbourhood and room type — a 30× gap. The cause is a single host, id 194230296, holding 28 of the 45 professional listings there, every one priced between R\$6,607 and R\$8,886. One operator flips an entire cell. This is a concrete, memorable demonstration of why the winsorisation choice in Finding 3 changes the answer, and it's the kind of detail that shows we actually looked at the data rather than at summary statistics.

4 Presenting this without looking like we're showing off

Order decides how it lands. Lead with the marketing conclusion (four competitive strategies), show the specification curve as the evidence, *then* name Silberzahn. Reversed, it reads as a statistics exercise in a marketing course.

Mechanically the curve is one chart: specifications sorted along the x-axis by effect size, effect on the y-axis, and a panel underneath marking which choices were active in each. Building it is a nested loop over the choice grid. There is nothing in it beyond Block V descriptive statistics, which is exactly why it works here — conceptually ahead of the room, technically inside the syllabus.

Presentation date. The deck freezes on 1 September for everyone regardless of when we present, so we cannot rewrite slides in response to other groups — but we can change what we *say*. Presenting on **11 September** means roughly six groups have gone first, several will have quoted a single confident professional-premium number, and the curve lands with context, while still leaving six days for the report. 16 September lands harder but leaves one day for the report. 2 September gives no context at all.

When we get there, the framing is not "group 4 was wrong." It's "this question has already been answered two different ways in this room, and that's exactly what should happen." Collegial, not combative. The point lands better and nobody gets defensive.

5 External data, which learning outcome 2 explicitly asks for

The outcome reads "use internal *and* external data." Every other group will stay inside the CSVs. Three cheap additions, well sourced, are enough:

- **Regulation.** Amsterdam's 30-night annual cap, LA's Home-Sharing Ordinance with its primary-residence requirement and day cap, Flanders' tourist accommodation registration, and Rio's comparatively light regime. This is what turns the availability spike (62% of Amsterdam listings at zero) from an observation into an explanation. Get the effective dates and cite them properly.
- **Housing stock or population per city** → listings per 1,000 dwellings. A penetration measure that cannot be computed from the files at all and is genuinely comparable across the four.
- **PPP conversion factors** (World Bank) for the currency robustness check.

Keep this to a handful of numbers. It's a supporting move, not a second project.

6 What to drop from the standard approach

- **Distance from the city centre.** The standard Airbnb analysis, and it's wrong here. Correlation with log price is *positive* in Rio (+0.15) and flat in LA (+0.08): Rio's priciest neighbourhoods sit 8–21 km from Centro because the value is the beach, and LA's is Malibu at 45 km. Antwerp has exactly one listing beyond 10 km. Use within-city neighbourhood rankings instead.
- **Estimated revenue per listing.** Needs an assumed review rate and an assumed stay length, neither in the data, applied to a base of 0.26–0.85 reviews per month. Appendix at most, clearly labelled.
- **Top-10 host share.** Antwerp's 10.8% against 2–3% elsewhere is arithmetic, not concentration — ten hosts are a bigger slice of 2,400 listings than of 33,000. Use share of listings held by hosts with 3+ (LA 39.2%, Antwerp 33.4%, Rio 26.8%, Amsterdam 9.7%).
- **Total review count as a demand measure.** It mostly measures how long a listing has existed.
- **Budget / mid / premium / luxury tiers.** Generic, and every deck will have them. Segment on what the data earned instead: room type × host type × whether the listing has ever been booked.
- **A 2×2 positioning map with proxy axes.** Looks like strategy, carries no information.

7 Mechanics

One person writes the cleaning script on day one and everyone imports it, because four people cleaning four files four ways will not reconcile. The rules: drop the 19 listings priced at zero; report medians, never means (Rio tops out at R\$132,358); cap minimum nights at 365 but keep the >30 group, it's a finding; fill `reviews_per_month` with 0 where it's blank, since it's blank exactly when review count is zero and dropping those rows would delete 42% of Rio; ignore `last_review` per the brief. Add two derived columns centrally: price index (price ÷ that city's median) and a professional flag.

Also worth stating early: these are city files, not country files. "Belgium" is Antwerp, "Netherlands" is Amsterdam's 22 districts, "California" is LA County. And `neighbourhood_group` is entirely empty except in LA, so level-1 analysis is LA-

only.

When	What
26–27 Aug	Cleaning script frozen. Agree the per-city summary table. One person starts the specification-curve loop — it's the long pole.
28–30 Aug	City analyses in parallel, on the four-principles spine. External data gathered and sourced.
31 Aug	Assemble, rehearse aloud once, and pressure-test Findings 3 and 4 by arguing against them.
1 Sep, 11:59pm	Deck uploaded, one person only.
11 Sep	Present.
12–17 Sep	Report, incorporating what the presentation surfaced.

8 How this could backfire

- **It reads as showing off.** Only if we lead with method. Lead with the four strategies.
- **Referencing other groups reads as arrogant.** Handled by framing it as expected variation rather than as anyone's error. Rehearse that sentence.
- **The professor may not care about his own reference list.** Possible. Every finding above stands on its own merits regardless, so the downside is small.
- **Rio will be challenged**, and by the person most likely to know Rio. Whoever presents it must be able to produce the within-neighbourhood control (56 paired cells, pros cheaper in 36) on demand, and should know the Catete example cold.
- **Over-claiming.** This is one 2023 cross-section with no policy variable in it. Regulation is *consistent with* the patterns; it did not *cause* them. No causal language anywhere, and a proper limitations slide: listed price isn't paid price, reviews only proxy bookings, availability is ambiguous between booked, blocked and capped, and there is nothing on property size or quality.

All figures computed directly from the four CSVs in `datasets/`; listings priced at zero excluded throughout. Finding 1 uses neighbourhoods with ≥ 30 listings of each room type; Finding 4 uses cells with ≥ 15 per side. Verify the exact figures in the Ehrenberg and Silberzahn papers before citing them in the report.